

Q1. A teacher has asked you to build a simple Number Inspector that analyzes a list of numbers submitted by students. For each number, the program should display all of its factors and determine whether it is a prime number. Write a program that takes an integer n and reads n integers one by one. For each number find all of its factors. If the number has exactly two factors, print "Prime". Otherwise, print "Not Prime". **[15 marks]**

Input	Output
4 10 7 15 13	Factors of 10: 1 2 5 10 Not Prime Factors of 7: 1 7 Prime Factors of 15: 1 3 5 15 Not Prime Factors of 13: 1 13 Prime

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            int num = sc.nextInt();
            int factorCount = 0;

            System.out.print("Factors of " + num + ": ");

            for (int j = 1; j <= num; j++) {
                if (num % j == 0) {
                    System.out.print(j + " ");
                    factorCount++;
                }
            }
            System.out.println();
            if (factorCount == 2) {
                System.out.println("Prime");
            } else {
                System.out.println("Not Prime");
            }
        }
    }
}
```

Q1. A digital lock accepts only special numbers as valid access codes. A number is considered special if every non-zero digit in the number evenly divides the number itself. Write a program that takes an integer n and reads n integers one by one. For each number, count how many digits it has and count how many of its digits evenly divide the number (ignore the digit 0). If every non-zero digit divides the number print “Special”, otherwise print “Not special”. **[15 marks]**

Input	Output
4 128 26 101 75	Special Number Not Special Special Number Not Special

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i = 1; i <= n; i++) {
            int num = sc.nextInt();
            int temp = num;
            int digitCount = 0;
            int divisibleCount = 0;

            while (temp > 0) {

                int digit = temp % 10;
                digitCount++;

                if (digit != 0 && num % digit == 0) {
                    divisibleCount++;
                }

                temp /= 10;
            }

            if (digitCount == divisibleCount) {
                System.out.println("Special Number");
            } else {
                System.out.println("Not Special");
            }
        }
    }
}
```

Q1. A data analysis system monitors numeric IDs and classifies them based on the balance of even and odd digits. The system continues processing IDs until the user enters -1. Write a program that keeps taking positive integers until -101 is entered. For each number, count how many even digits and odd digits it contains. The program should print "More Even Digits" if the number contains more even digits than odd digits; "More Odd Digits" if the number contains more odd digits than even digits; "Equal Even and Odd Digits" if the counts are equal. **[15 marks]**

Input	Output
248 1357 1234 90 -101	More Even Digits More Odd Digits Equal Even and Odd Digits Equal Even and Odd Digits

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int num = sc.nextInt();
        while (num != -101) {
            int even = 0;
            int odd = 0;
            int temp = num;
            while (temp > 0) {
                int digit = temp % 10;
                if (digit % 2 == 0) {
                    even++;
                } else {
                    odd++;
                }
                temp /= 10;
            }
            if (even > odd) {
                System.out.println("More Even Digits");
            } else if (odd > even) {
                System.out.println("More Odd Digits");
            } else {
                System.out.println("Equal Even and Odd Digits");
            }
            num = sc.nextInt();
        }
    }
}
```

Q1. A statistics program records a sequence of integers entered by the user. The program continues reading numbers until the sentinel value 999 is entered. It then displays a summary of the numbers entered. Write a program that keeps taking integers until 999 is entered. The program should count how many numbers are Positive (consider 0 as positive), Negative, Even, Odd. The program should not include 999 in any of the counts and should print the totals after the input ends. **[15 marks]**

Input	Output
5 -2 0 11 8 -7 999	Positive: 4 Negative: 2 Even: 3 Odd: 3

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int positive = 0;
        int negative = 0;
        int even = 0;
        int odd = 0;
        int num = sc.nextInt();
        while (num != 999) {
            if (num >= 0) {
                positive++;
            } else {
                negative++;
            }
            if (num % 2 == 0) {
                even++;
            } else {
                odd++;
            }
            num = sc.nextInt();
        }

        System.out.println("Positive: " + positive);
        System.out.println("Negative: " + negative);
        System.out.println("Even: " + even);
        System.out.println("Odd: " + odd);
    }
}
```